

### **i) Brand or Product Name**

LORICAN (25% Tilmicosin Oral Solution)

### **ii) Name and Strength of Active Substance(s)**

Tilmicosin, 25% w/v

### **iii) Product Description**

A clear solution with the color of light yellow to yellow.

### **iv) Pharmacodynamics/Pharmacokinetics/Environmental Properties**

#### **Pharmacodynamics**

Tilmicosin is a semi-synthetic antibiotic of the macrolide group and is believed to affect protein synthesis. It has bacteriostatic action but at high concentrations it may be bactericidal. This antibacterial activity is predominantly against Gram-positive microorganism with activity against certain gram-negative ones and Mycoplasma of a bovine, porcine, ovine and avian origin. In particular, its activity has been demonstrated against the following microorganism:

*Pigs: Mycoplasma hyopneumoniae, Pasteurella multocida and Actinobacillus pleuropneumoniae*

*Chickens and turkeys: Mycoplasma gallisepticum and Mycoplasma synoviae*

*Calves: Mannheimia haemolytica, Pasteurella multocida, Mycoplasma bovis and M. dispar.*

Scientific evidence suggests that macrolides act synergistically with the host immune system. Macrolides appear to enhance phagocyte killing of bacteria. Tilmicosin has been shown to inhibit *in vitro* the replication of the Porcine Reproductive and Respiratory Syndrome virus in alveolar macrophages in a dose dependent fashion.

Cross-resistance between tilmicosin and other macrolides and lincomycin has been observed.

#### **Pharmacokinetics**

Whilst blood concentrations of tilmicosin are low, there is pH-dependent macrophage accumulation of tilmicosin in inflamed tissues.

*Pigs:* After oral administration of 200 mg tilmicosin/l drinking water, the average active substance concentrations detected in lung tissue, alveolar macrophages and bronchial epithelium 5 days after the start of treatment were 1.44 µg/mL, 3.8 µg/mL and 7.4 µg/g respectively.

Poultry: As early as 6 hours after oral administration of 75 mg tilmicosin/l drinking water, the average active substance concentrations detected in lung and alveolar tissue were 0.63 µg/g and 0.30 µg/g respectively. 48 hours after the start of treatment, the tilmicosin concentrations in lung and alveolar tissue were 2.3 µg/g and 3.29 µg/g respectively.

Calves: As early as 6 hours after oral administration of 25 mg tilmicosin/kg body weight/day in milk replacer, an average active substance concentration of 3.1 µg/g was detected in lung tissue. 78 hours after the start of treatment, the tilmicosin concentration in lung tissue was 42.7 µg/g. Therapeutically effective concentrations of tilmicosin were measured up to 60 hours after treatment.

Turkeys: After oral administration of 75 mg tilmicosin/l drinking water, the average active substance concentrations detected in lung tissue, air sac tissue and plasma 5 days after the start of treatment were 1.89 µg/mL, 3.71 µg/mL and 0.02 µg/g respectively. The highest mean tilmicosin concentration detected for lung tissues was 2.19 µg/g at 6 days; for air sac tissue it was 4.18 µg/g at 2 days and in the plasma it was 0.172 µg/g at 3 days.

## **v) Indication**

Pigs: For the treatment and prevention of respiratory disease in pig herds, associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida*, *Actinobacillus pleuropneumoniae* and other organisms susceptible to tilmicosin.

Chickens: For the treatment and prevention of respiratory disease in chicken flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Turkeys: For the treatment and prevention of respiratory disease in turkey flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Calves: For the treatment and prevention of bovine respiratory disease, associated with *Mannheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma bovis*, *M. dispar* and other organisms susceptible to tilmicosin.

## **vi) Recommended Dosage**

Pigs: To be included in the drinking water to provide a daily dose of 15-20 mg/kg bodyweight for 5 days, which may be achieved by the inclusion of 200 mg tilmicosin per L (80 mL LORICAN per 100 L).

Chickens and Turkeys (except hens producing eggs for human consumption): To be included in the drinking water at a daily dose of 15-20 mg/kg bodyweight in chickens and 10-27 mg/kg bodyweight in turkeys for 3 days, which may be achieved by the inclusion of 75 mg tilmicosin per L (30 mL LORICAN per 100 L).

Calves: To be included in milk replacer only, at a dose of 12.5 mg/kg bodyweight and given twice daily for 3-5 consecutive days, which may be achieved by the

inclusion of 1 mL of product every 20 kg bodyweight.

### **vii) Mode of Administration**

Oral via drinking water or milk replacer.

### **viii) Contraindications**

- \* Do not mix with other medicines.
- \* Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.
- \* Do not administer to layers the eggs of which are intended for human consumption.
- \* Treated poultry must not be slaughtered for human consumption during treatment.
- \* People handling the product must wear personal protective devices such as protective gloves of impermeable rubber and safety goggles and/or face shield. Do not eat or smoke in the area of material storage. Wash the hands with soap before eating or smoking.

### **ix) Warnings and Precautions**

- \* Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin-related substances. The use of the product should be based on susceptibility tests.
- \* Official, national and regional antimicrobial policies should be taken into account when the product is used.
- \* Severely ill individuals tend to drink less and may need simultaneous treatment, preferably by parenteral medication.

### **x) Interactions with Other Medicaments**

In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other products.

### **xi) Statement on usage during pregnancy and lactation**

In the absence of toxicology study, this veterinary medicinal product must not be used during pregnancy and lactation. Do not use in laying birds producing eggs for human consumption.

## **xii) Adverse Effects/Undesirable Effects**

In very rare cases, a decrease in water intake has been observed.

## **xiii) Overdose and Treatment**

No symptoms of overdose have been seen in chickens given drinking water containing levels of tilmicosin up to 375 mg/L (equivalent to 75-100 mg/kg bodyweight or 5 times the recommended dose) for 5 days; daily treatment with 75 mg/L (equivalent to the maximum recommended dose) for 10 days resulted in a reduction in faecal consistency.

No symptoms of overdose have been seen in turkeys given drinking water containing levels of tilmicosin up to 375 mg/L (equivalent to 50-135 mg/kg bodyweight or 5 times therecommended dose) for 3 days; daily treatment with 75 mg/L (equivalent to the maximum recommended dose) for 6 days also produced no symptoms of overdose.

## **xiv) Withdrawal Period(s)**

Pigs - 14 days

Chickens-12 days

Turkeys – 19 days

Calves - 42 days

Not authorised for use in laying birds producing eggs for human consumption.

Do not use within 14 days of onset of the laying.

Not for use in lactating animals.

## **xv) Storage Conditions**

Store below 25 degree Celsius.

## **xvi) Shelf Life**

Shelf Life: 2 years.

Shelf-life after First Opening the Container: 3 months.

Shelf Life after Dilution: it's mixed with water for directly use. And the medicated water should be used in 24hs.

**xvii) Dosage Forms and packaging available**

Dosage form: Solution for oral administration

Pack size: 1 L\*10 bottles/carton

**xvi) Name and Address of Manufacturer/ Product Registration Holder****Manufacturer:**

QILU ANIMAL HEALTH PRODUCTS CO., LTD.

23999, Gongye North Road, Licheng District, Jinan,  
Shandong, 250100 China.

**Product Registration Holder:**

RURAL CHEMICAL INDUSTRIES SDN BHD

5 & 7, Lorong SS 13/6A, Subang Jaya Industrial Estate,  
47500 Subang Jaya, Selangor, Malaysia.

**xvii) Date of Revision of Package Insert**

24 April 2024